

Foglio 1 - Soluzioni,

ESERCIZIO 1

Mostrare che $D = \{(x, y) \in \mathbb{R}^2 \mid x^4 + y^4 + 2x^2y \leq 4\}$ non è y -normale.

Soluzione

D è simmetrico rispetto all'asse y , i.e. $(x, y) \in D \iff (-x, y) \in D$.

Dunque, se D fosse y -normale, avrei che

$$\exists \varphi: [a, b] \rightarrow \mathbb{R}, \varphi \in C^0([a, b]), \text{ t.c. } D = \{(x, y) \in \mathbb{R}^2 \mid a \leq y \leq b, -\varphi(y) \leq x \leq \varphi(y)\}$$

Costruiamo che $\nexists$ una tale φ . Lavoriamo su $\{x \geq 0\}$.

Troviamo a e b .

$$\begin{cases} x^4 + y^4 + 2x^2y = 4 \\ x = 0 \end{cases} \longrightarrow y_{1,2} = \pm\sqrt{2} \longrightarrow [a, b] = [-\sqrt{2}, \sqrt{2}].$$

$$x^4 + y^4 + 2x^2y = 4 \iff x^4 + 2yx^2 + (y^4 - 4) = 0$$

$$\iff x^2 = -y + \sqrt{y^2 - y^4 + 4}$$

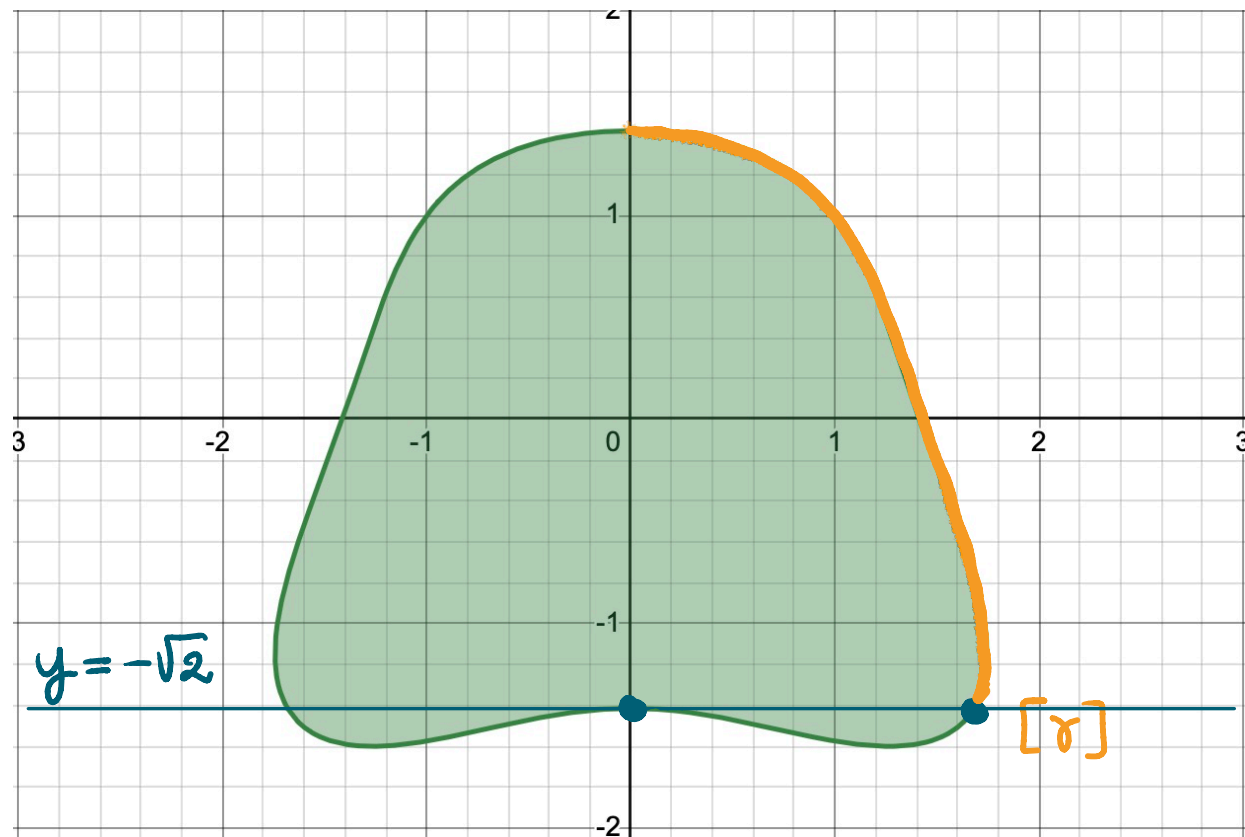
$$x > 0 \rightsquigarrow x = \sqrt{\sqrt{y^2 - y^4 + 4} - y}$$

Se D fosse y -normale, avrei

$$\partial D \cap \{x \geq 0\} = [\gamma] \quad \text{con} \quad \gamma: [-\sqrt{2}, \sqrt{2}] \longrightarrow \mathbb{R}^2$$
$$t \longmapsto (\sqrt{t^2 - t^4 + 4} - t, t)$$

$\longrightarrow \gamma(-\sqrt{2}) = (\sqrt[4]{8}, -\sqrt{2}) \in \partial D \cap \{x \geq 0\}$
Ma anche $(0, -\sqrt{2}) \in \partial D \cap \{x \geq 0\}$ $\rightarrow$ 2 valori con la stessa y !

Dunque $[\gamma]$ non può essere $\partial D \cap \{x \geq 0\}$

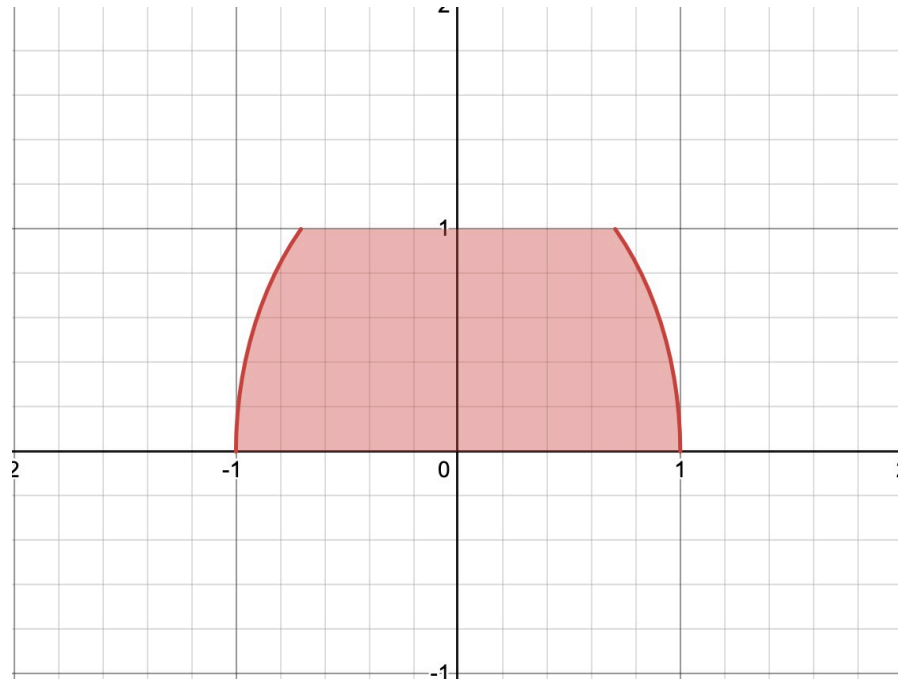


ESERCIZIO 2

Esprimere $E = \{(x, y) \in \mathbb{R}^2 \mid 2x^2 + y^2 \leq 2, 0 \leq y \leq 1\}$ come dominio x e y -normale (se possibile)

Soluzione

$$2x^2 + y^2 \leq 2 \rightarrow x^2 + \frac{y^2}{2} \leq 1 : \text{ellisse semiasse } 1 \text{ e } \sqrt{2}$$



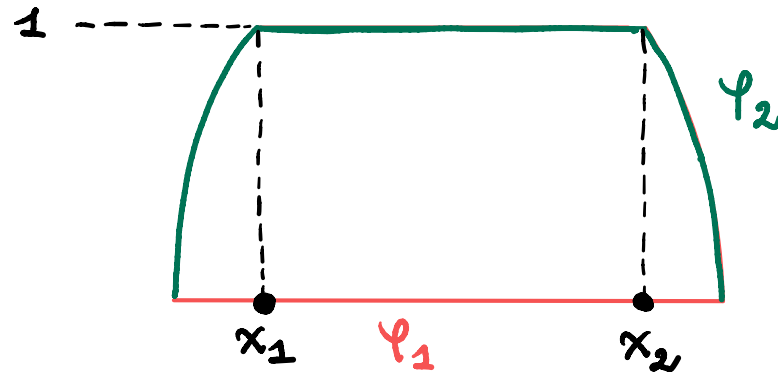
È evidente la x -normalità e y -normalità.

Scrittura y -normale: $0 \leq y \leq 1$ e $x^2 \leq 1 - y^2/2$ i.e. $-\sqrt{1 - y^2/2} \leq x \leq \sqrt{1 - y^2/2}$

$$E = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq y \leq 1, -\sqrt{1 - y^2/2} \leq x \leq \sqrt{1 - y^2/2}\}$$

Scrittura x -normale: Charamente $-1 \leq x \leq 1$. Voglio

$$\varphi_1(x) \leq y \leq \varphi_2(x)$$



$$\varphi_1(x) = 0$$

$\varphi_2(x)$ è definita a tratti. Ci servono x_1 e x_2 .

$$\begin{cases} y=1 \\ 2x^2+y^2=2 \end{cases} \Rightarrow x_{1,2} = \pm \frac{1}{\sqrt{2}}$$

Nelle parti "curve", $\varphi_2(x)$ è $\sqrt{2(1-x^2)}$, dunque usando tutto

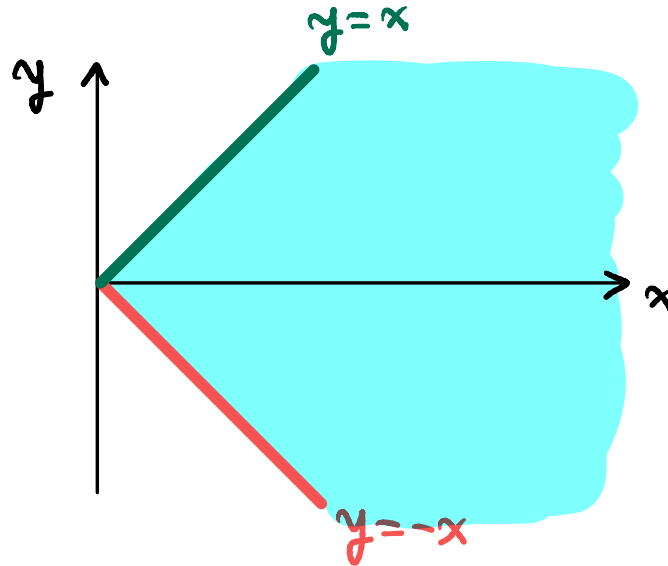
$$\varphi_2(x) = \begin{cases} \sqrt{2(1-x^2)} & 1/\sqrt{2} < |x| < 1 \\ 1 & |x| < 1/\sqrt{2} \end{cases}$$

ESERCIZIO 3

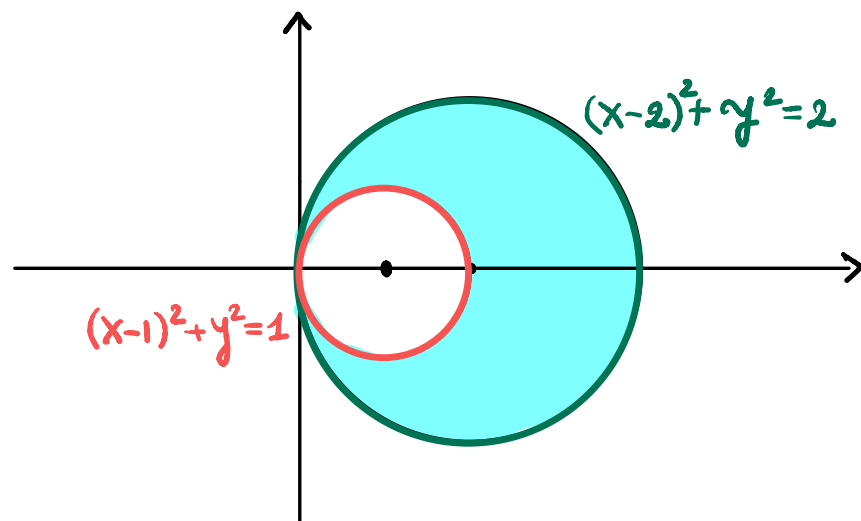
Esprimere $F = \{(x,y) \in \mathbb{R}^2 \mid 2x \leq x^2 + y^2 \leq 4x, |y| \leq x\}$ come dominio x e y -normale (se possibile)

Soluzione

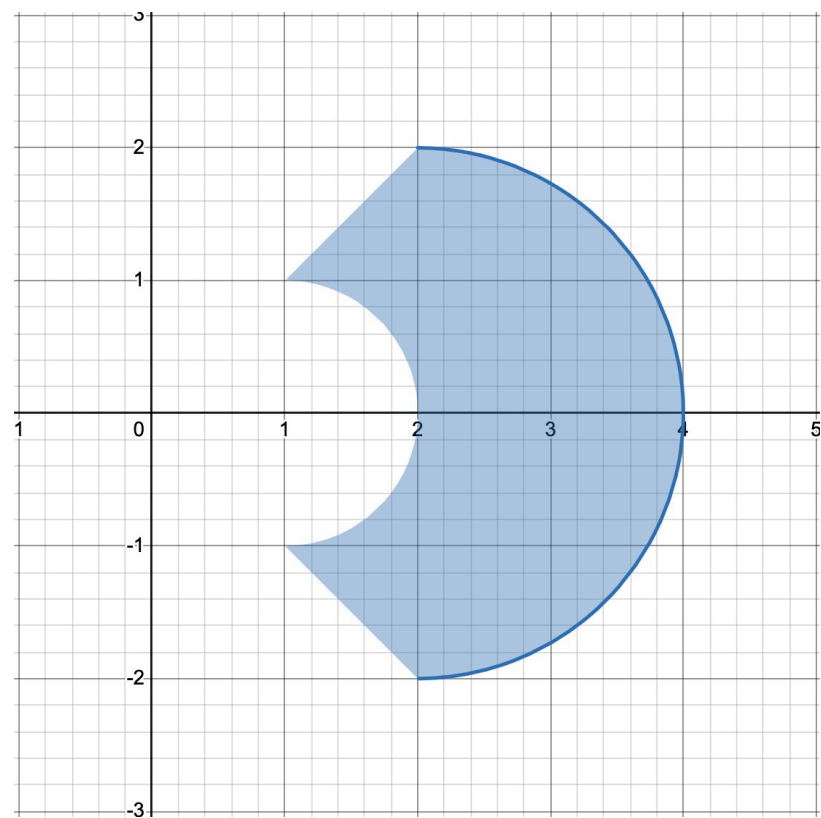
Disegniamo F . $|y| \leq x \rightarrow -x \leq y \leq x$



$$\begin{aligned} 2x \leq x^2 + y^2 \leq 4x &\begin{cases} \nearrow x^2 + y^2 \leq 4x \Rightarrow x^2 - 4x + 4 - 4 + y^2 \leq 0 \Rightarrow (x-2)^2 + y^2 \leq 4 \\ \searrow x^2 + y^2 \geq 2x \Rightarrow x^2 - 2x + 1 - 1 + y^2 \geq 0 \Rightarrow (x-1)^2 + y^2 \geq 1 \end{cases} \\ &\begin{matrix} \text{cerchio centro } (2,0) \text{ raggio } 2 \\ \text{fuori dal cerchio centro } (1,0) \\ \text{raggio } 1 \end{matrix} \end{aligned}$$

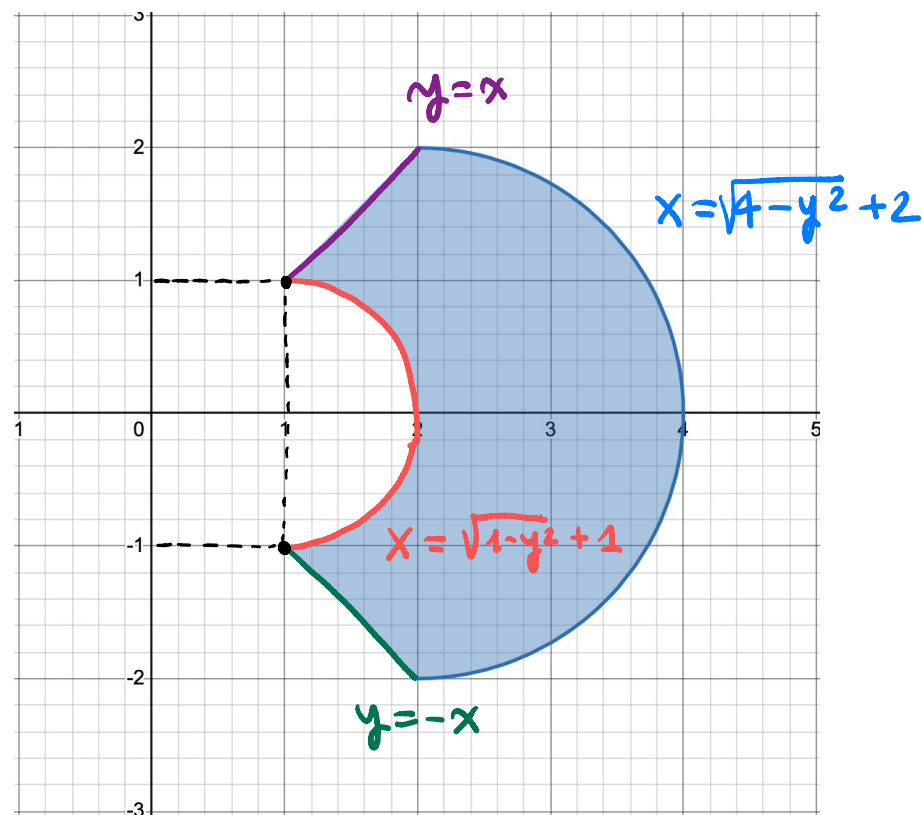


Intersecando:



Si vede immediatamente la non x -normalità.

Per la y -normalità, abbiamo $-2 \leq y \leq 2$. Poi $\varphi_1(y) \leq x \leq \varphi_2(y)$



$$\varphi_2(y) = \sqrt{4-y^2} + 2$$

$$\varphi_1(y) = \begin{cases} 1 + \sqrt{1-y^2} & |y| < 1 \\ |x| & 1 \leq |y| \leq 2 \end{cases}$$

$$\Rightarrow D = \{(x,y) \in \mathbb{R}^2 \mid -2 \leq y \leq 2, \varphi_1(y) \leq x \leq \varphi_2(y)\}$$

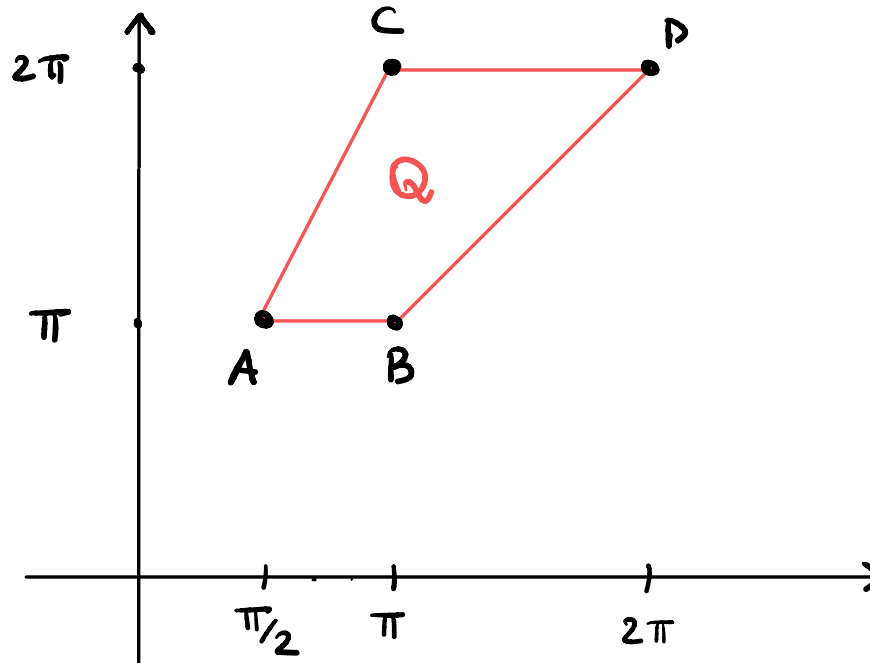
ESERCIZIO 4

Calcolare

$$\iint_Q \frac{\sin y}{y} dx dy$$

con Q quadrilatero di vertici $(\pi/2, \pi), (\pi, \pi), (\pi, 2\pi), (2\pi, 2\pi)$

Soluzione



È comodo vedere Q come y -normale, dove $\pi \leq y \leq 2\pi$.

x è delimitato dalle rette $\overline{AC} \rightarrow x=y$

$$\overline{BD} \rightarrow x=y/2$$

$$\text{Dunque} \quad \iint_Q \frac{\sin y}{y} dx dy = \int_{\pi}^{2\pi} \left(\int_{y/2}^y \frac{\sin y}{y} dx \right) dy$$

$$\begin{aligned}
 &= \int_{\pi}^{2\pi} \frac{\sin y}{y} \left(y - \frac{y}{2} \right) dy \\
 &= \frac{1}{2} \left[-\cos(2\pi) + \cos(\pi) \right] = 1.
 \end{aligned}$$

Nota: Provate a farlo vedendo Q come x -normale.

Dovrete integrare $\int \frac{\sin y}{y} dy \dots$ buona fortuna! 😊

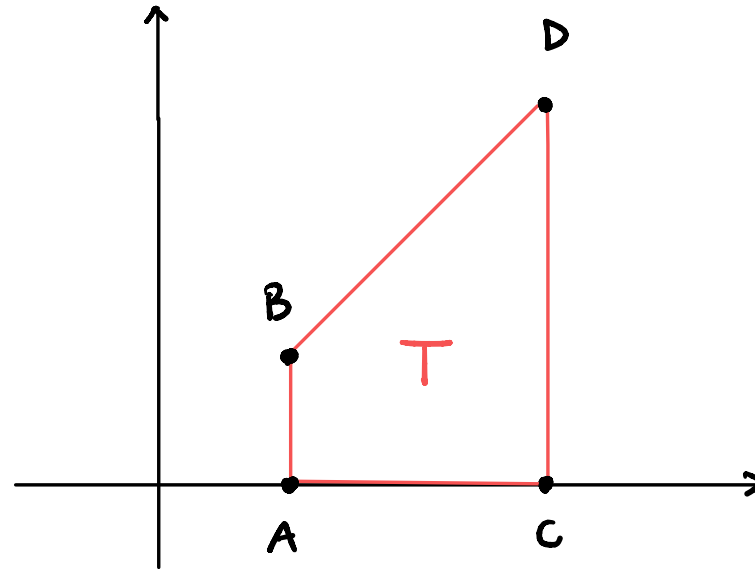
ESERCIZIO 5

Calcolare

$$\iint_T \frac{dx dy}{x^2 + y^2}$$

dove T è il trapezio di vertici $(1,0)$, $(1,1)$, $(3,0)$, $(3,3)$.

Soluzione



Vedendo T come dominio x -normale, ho

$$T = \{(x, y) \in \mathbb{R}^2 \mid 1 \leq x \leq 3, 0 \leq y \leq x\}$$

Dunque:

$$\begin{aligned} \iint_T \frac{dx dy}{x^2 + y^2} &= \int_1^3 \left(\int_0^x \frac{1}{x^2 + y^2} dy \right) dx \\ &= \int_1^3 \frac{1}{x^2} \left(\int_0^x \frac{1}{1 + (y/x)^2} dy \right) dx \end{aligned}$$

$$\begin{aligned}
 &= \int_1^3 \frac{1}{x} \left[\arctan\left(\frac{y}{x}\right) \right]_0^x dx \\
 &= \frac{\pi}{4} \int_1^3 \frac{1}{x} dx \\
 &= \log 3 \cdot \pi/4
 \end{aligned}$$

ESERCIZIO 6

Calcolare

$$\iint_D \frac{x^3 e^{-y^2}}{x^2 + y^2} dx dy$$

dove D è il dominio dell' ES. 1.

Soluzione

$$\text{Sia } f(x, y) = \frac{x^3 e^{-y^2}}{x^2 + y^2}$$

$$\left. \begin{array}{l} \bullet f(x, y) = -f(-x, y) \\ \bullet (x, y) \in D \iff (-x, y) \in D \end{array} \right\} \implies \iint_D f(x, y) dx dy = 0.$$

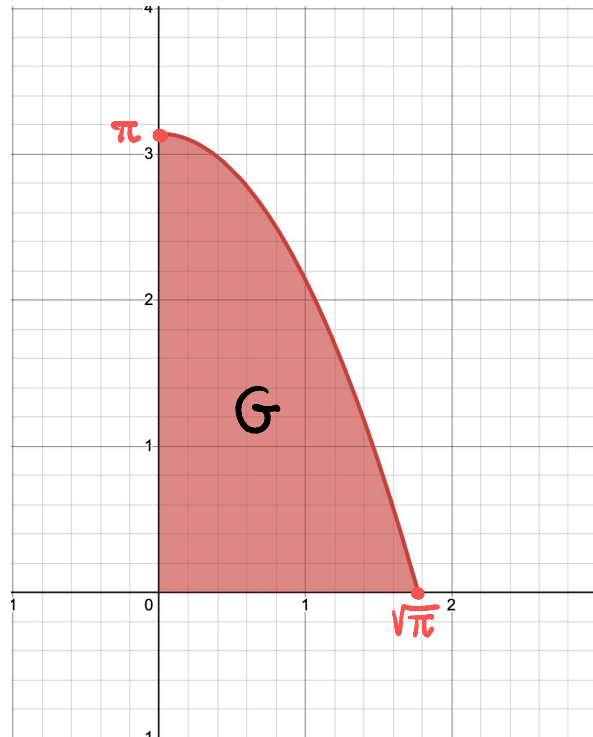
ESERCIZIO 7

Calcolare

$$\iint_G x^3 \sin^2(x^2) dx dy$$

dove $G = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y \leq \pi, x \geq 0, y \geq 0\}$

Soluzione



G è ovviamente y -normale, con

$$G = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq y \leq \pi, 0 \leq x \leq \sqrt{\pi - y}\}$$

Dunque

$$\iint_G x^3 \sin^2(x^2) dx dy = \int_0^\pi \left(\int_0^{\sqrt{\pi-y}} x^3 \sin^2(x^2 - \pi) dx \right) dy$$

periodo $\sin^2 z = \pi$

$$= \frac{1}{2} \int_0^\pi \left(\int_{-\pi}^{-y} (u + \pi) \sin^2 u du \right) dy$$

$u = x^2 - \pi$
 $du = 2x dx$
 $x^2 = u + \pi$

$$= \frac{1}{2} \int_0^\pi \left(\pi \int_{-\pi}^{-y} \sin^2 u du + \int_{-\pi}^{-y} u \sin^2 u du \right) dy = (*)$$

Calcoliamo a parte: bisezione

$$\begin{aligned} \bullet) \int_{-\pi}^{-y} \sin^2 u du &= \int_{-\pi}^{-y} \left(\frac{1}{2} - \frac{\cos 2u}{2} \right) du \\ &= \frac{1}{2} [-y + \pi] - \frac{1}{4} [\sin 2u]_{-\pi}^{-y} \\ &= -\frac{1}{2} y + \frac{\pi}{2} + \frac{1}{4} \sin 2y \end{aligned}$$

$$\begin{aligned} \bullet) \int_{-\pi}^{-y} u \sin^2 u du &= \int_{-\pi}^{-y} \left(\frac{u}{2} - \frac{u \cos 2u}{2} \right) du \\ &= \frac{1}{4} [y^2 - \pi^2] - \frac{1}{2} \int_{-\pi}^{-y} u \cos 2u du \end{aligned}$$

bisezione

per parti

$$\begin{aligned}
 &= \left| \frac{1}{4} (y^2 - \pi^2) - \frac{1}{8} [2y \sin(2y) + \cos(2y)] \right|_{\pi}^y \\
 &= \frac{1}{4} (y^2 - \pi^2) - \frac{1}{8} (2y \sin(2y) + \cos(2y) - 1)
 \end{aligned}$$

Ritorniamo a (*).

$$\begin{aligned}
 (*) &= \frac{1}{2} \int_0^{\pi} \left[-\frac{\pi}{2} y + \frac{\pi^2}{2} + \frac{\pi}{4} \sin 2y + \frac{1}{4} (y^2 - \pi^2) - \frac{1}{8} (2y \sin(2y) + \cos(2y) - 1) \right] dy \\
 &= \frac{1}{2} \left[-\frac{\pi}{4} y^2 + \frac{\pi^2}{2} y - \frac{\pi}{8} \cos 2y + \frac{y^3}{12} - \frac{\pi^2}{4} y + \frac{1}{16} \sin(2y) + \frac{1}{8} y \right]_0^{\pi} - \frac{1}{8} \int_0^{\pi} y \sin(2y) dy \\
 &= \frac{1}{2} \left[\cancel{-\frac{\pi^3}{4}} + \cancel{\frac{\pi^3}{2}} - \cancel{\frac{\pi}{8}} + \frac{\pi^3}{12} - \cancel{\frac{\pi^3}{4}} + \cancel{\frac{\pi}{8}} + \frac{\pi}{8} \right] - \frac{1}{32} [\sin(2y) - 2y \cos(2y)]_0^{\pi} \\
 &\stackrel{\text{parti su } y \sin(2y) dy}{=} \frac{1}{2} \left[\frac{\pi^3}{12} + \frac{\pi}{8} \right] + \frac{\pi}{16} \\
 &= \frac{\pi^3}{24} + \frac{\pi}{16} + \frac{\pi}{16} = \frac{\pi^3}{24} + \frac{\pi}{8}
 \end{aligned}$$